

Day 6 – Cyber Security & Ethical Hacking

Wayback Machine in Cyber Crime Investigation

This day focuses on **digital footprints**, **evidence recovery**, and **OSINT techniques** used in cybercrime investigations.

Key Concept: Digital Footprints Never Truly Disappear

Many scammers believe:

- ✗ Deleting a website
- ✗ Taking down hosting
- ✗ Removing social media pages

...will erase all evidence.

Reality:

The internet **remembers everything** through:

- Web archives
- Search engine caches
- Screenshots
- Third-party data collectors

The **Wayback Machine** is one of the most powerful tools proving this fact.

What is the Wayback Machine?

Definition

The **Wayback Machine** is a **web archiving service** created by the **Internet Archive** that stores historical snapshots of websites over time.

It allows investigators to:

- View deleted websites
- Track changes in content
- Recover scam offers and fake claims
- Preserve digital evidence

Website: <https://archive.org/web>

How the Wayback Machine Works (Technical Insight)

1. **Web crawlers** periodically visit websites
 2. Pages are **snapshotted** (HTML, images, text)
 3. Each snapshot is stored with:
 - Date
 - Time
 - URL
 4. Even if the site is later deleted, **archived copies remain**
 - Archives are **read-only** → content cannot be altered later
 - This makes them **credible evidence**
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Attacker vs Investigator Mindset

Scammer Thinks

Website deleted

Domain expired

Victims have no proof

Identity hidden

Investigator Knows

Archived copies exist

Content still retrievable

Digital evidence preserved

Patterns can be reconstructed

Day 6 Practical Investigation Workflow

Step 1: Identify the Scam Domain

Example:

abcd.com

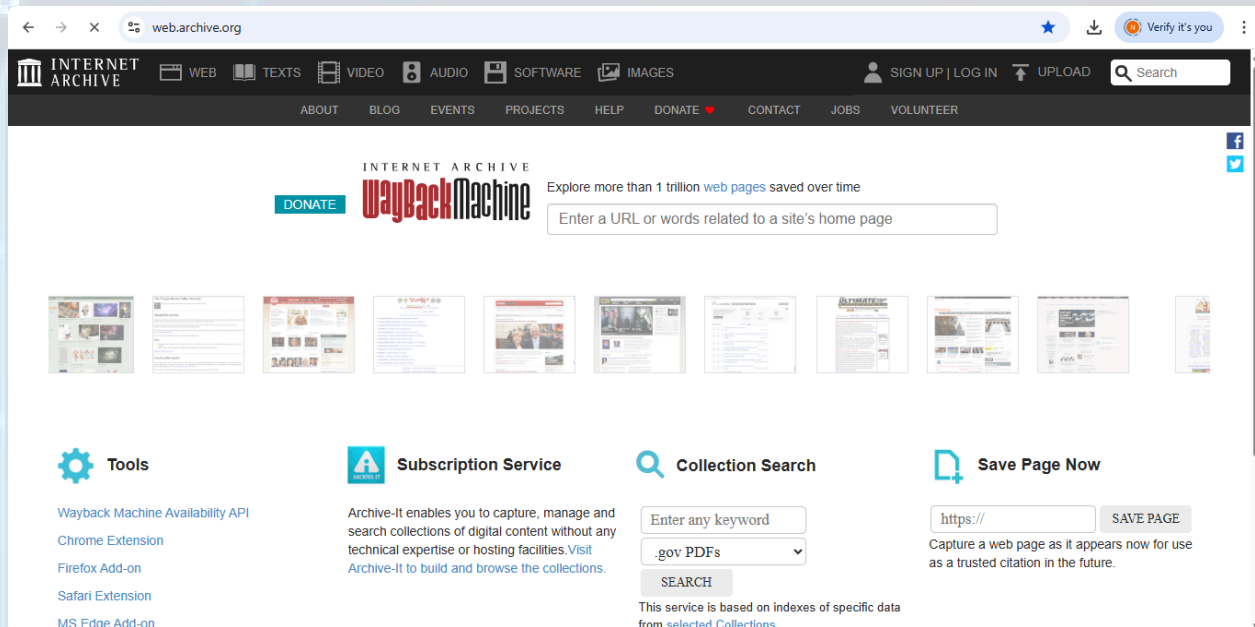
The website is currently:

-  Offline
-  Domain expired
-  Hosting removed

Step 2: Search in Wayback Machine

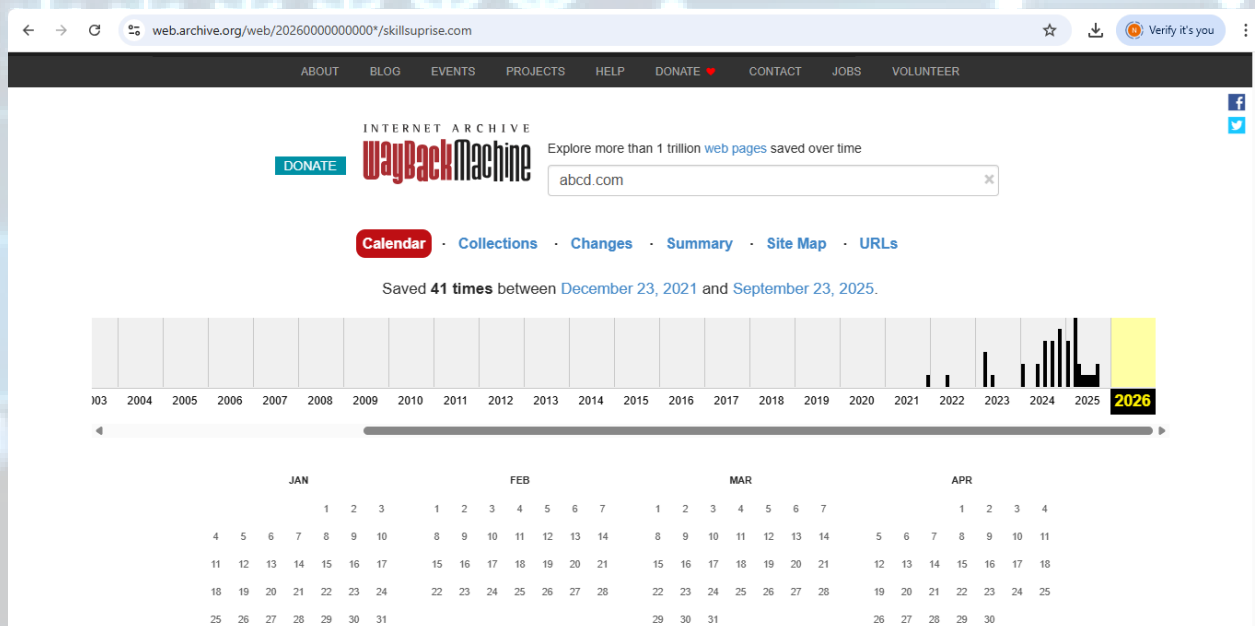
<https://web.archive.org/>

Enter:



Ex : abcd.com

Step 3: Analyze Timeline (Snapshots) : You may see:

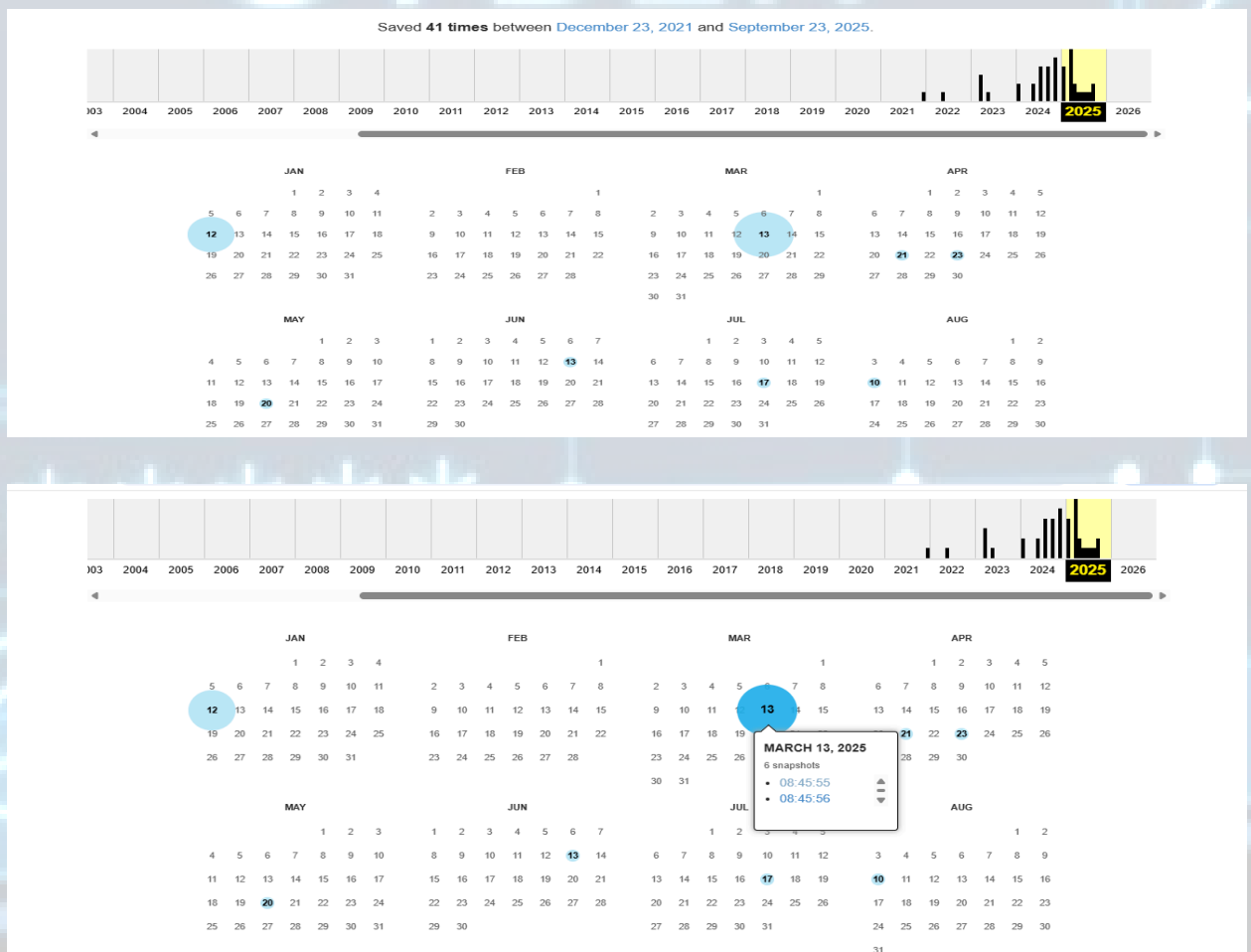


Day 6_100 DAYS CHALLENGE

2019 → 4 snapshots
2020 → 28 snapshots
2021 → 15 snapshots
2025 → 2 snapshots

This timeline shows:

- When the scam started
- Peak scam activity
- When the site vanished



Recovered Digital Evidence (Deep Analysis)

1 Fake Offers Page

Recovered Content:

"Invest ₹10,000 and earn ₹1,00,000 in 7 days - 100% guaranteed!"

🔍 Insight:

- Unrealistic ROI
- Guaranteed profit claim
- Clear financial fraud indicator

2 Fake Testimonials

Recovered HTML:

"I earned ₹5 lakhs in one month. This platform changed my life!"
- Ravi Kumar, Hyderabad

🔍 Investigation Findings:

- Images are **stock photos**
- Same testimonial reused on multiple scam sites
- Names are fabricated

3 Contact & Payment Details

Recovered Data:

Email: support@abcd.com

Telegram: @FastMoneySupport

UPI ID: fastmoney@upi

 OSINT Value:

- Telegram handle traceable
- Email reused across scam networks
- UPI ID links to financial mule accounts

Hidden Pages (Often Overlooked)

Recovered URLs:

`/admin-login`
`/terms.html`
`/refund-policy.html`

 Why This Matters:

- Shows scammer's operational structure
- Fake legal pages expose **intent to deceive**
- Admin panels confirm **organized fraud**

Conclusion:

Website deletion was an attempt to evade law enforcement.
Archived data proves intent, operation period, and fraud patterns.

Legal & Ethical Importance

Wayback Machine evidence is used by:

- 🕵️ Cyber crime cells
- 🏛️ Law enforcement agencies
- 🏛️ Courts (supporting evidence)
- 🛡️ OSINT investigators

📌 It **does not** hack websites

📌 It retrieves **publicly archived data**

Core Lesson – Day 6 Takeaway

Deleting a website does not delete history.

The internet leaves **permanent footprints**, and OSINT tools like the Wayback Machine transform those footprints into **digital evidence**